

Software Design Description (SDD)

AI-Based Smart HVAC Controller

1. System Architecture

The system consists of Input Processing, AI Controller, Stateflow HVAC Logic and Actuator Control layers. Sensor inputs are processed and passed to the AI predictor and Stateflow controller.

2. Input Processing Design

Input signals T_Cabin, T_Ambient, T_Setpoint, Humidity_sensor and Q_Solar are validated. Temp_Error, High_Humidity_Flag and High_Solar_Flag are generated.

3. Temperature Processing

$\text{Temp_Error} = \text{T_Setpoint} - \text{T_Cabin}$. Positive values indicate heating demand and negative values indicate cooling demand.

4. Humidity Processing

Humidity sensor input is monitored. When humidity exceeds the threshold, High_Humidity_Flag is asserted to support Defog mode.

5. Solar Compensation Processing

Q_Solar is monitored. High_Solar_Flag is generated when solar load exceeds the configured threshold.

6. AI Controller Design

A MATLAB Function block predicts HVAC operating mode using T_Cabin, T_Setpoint, Humidity_sensor, Q_Solar and T_Ambient. The AI output is AI_HVAC_State.

7. AI State Mapping

1 = IDLE, 2 = COOLING, 3 = HIGH_COOLING, 4 = HEATING, 5 = DEFOG.

8. Stateflow Controller Design

The Stateflow chart contains OFF, IDLE, COOLING, HIGH_COOLING, HEATING and DEFOG states. Transitions are driven by HVAC inputs and AI_HVAC_State.

9. OFF State Design

All outputs disabled. Compressor, Heater and Blower commands are reset.

10. IDLE State Design

Maintains comfort state when no cooling, heating or defog action is required.

11. COOLING State Design

Compressor enabled, medium blower speed and face airflow mode activated.

12. HIGH_COOLING State Design

Maximum cooling demand. Compressor ON and blower speed set to maximum.

13. HEATING State Design

Heater enabled with foot airflow mode and medium blower speed.

14. DEFOG State Design

Compressor and heater enabled together to remove windshield fog.

15. Interface Design

Inputs: IGNITION_ON_OFF, T_Cabin, T_Ambient, T_Setpoint, Humidity_sensor, Q_Solar, HVAC_MODE_REQ. Outputs: Heater_Command, Compressor_Command, Blower_Speed, Recirc_Command, Airflow_Mode, AI_HVAC_State.

16. Verification Strategy

MIL testing is performed using Simulink Test Manager. Baseline testing and functional test cases are executed.

17. Design Assumptions

Valid sensor inputs, calibrated thresholds and trained AI model are assumed.

18. Future Enhancements

Deep learning integration, AUTOSAR support, code generation and HIL testing.